

2022

Time : 3 hours

Full Marks : 40

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

(Compulsory)

1. Pick up the correct alternative for each of
the following questions : 1×10=10

(a) If $y = e^{mx}$, then y_n is equal to :

(i) m

(ii) n

~~(iii)~~ $m^n e^{mx}$

(iv) $n^m e^{nx}$

(b) If $y = \sin(ax + b)$ then y_n is equal to :

(i) $\sin(ax + b)$

~~(ii)~~ $a^n \sin\left(ax + b + \frac{n\pi}{2}\right)$

(iii) $a^n \sin(ax + b + n\pi)$

(iv) None of these

(c) The maximum value of $\cos x$ in $\mathbb{R}$ is :

(i) 0

(ii) -1

~~(iii)~~ 1

(iv) None of these

(d) If u be a function of x and y then

$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$ is equal to :

☒ (i) du

(ii) df

(iii) ∂u

(iv) None of these

(e) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ is equal to :

(i) $\frac{\pi}{2}$

☒ (ii) $\frac{\pi}{4}$

(iii) π

(iv) None of these

(f) The area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :

☒ (i) $\pi^2 a^2 b$

☒ (ii) πab

(iii) a^2b

(iv) None of these

(g) $\int_0^{\infty} \frac{1}{1+x^2} dx$ is equal to :

(i) 0

(ii) ∞

~~(iii) $\frac{\pi}{2}$~~

(iv) None of these

$$\begin{aligned} & \left(\tan^{-1} x \right) \Big|_0^{\infty} \\ & \tan^{-1} \infty - \tan^{-1} 0 \\ & = \frac{\pi}{2} - 0 \end{aligned}$$

(h) The number of arbitrary constants in the general solution of linear differential equation of second order is/are :

(i) 0 (ii) 1

~~(iii) 2~~ (iv) 3

(i) Integrating factor of the differential

equation $\frac{dy}{dx} + y \cot x = 2 \cos x$ is :

- (i) $\sin x$
(ii) $\cos x$
(iii) $\sec x$
(iv) $\tan x$
- (j) The general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$ is :
- (i) $\log y = kx$
(ii) $ny = k$
(iii) $y = kx$
(iv) $y = k \log x$

Section-B

(Short answer type questions)

Answer any **three** questions of the following :

$$5 \times 3 = 15$$

2. State and prove Leibnitz theorem.

3. State and prove Maclaurin theorem.

4. Find the maximum and minimum value of $3x^3 - 7x^2 + 15x - 10$.
5. Explain the method of finding the solution of linear differential equation of first order.
6. Explain method of finding the complementary function (CF) of a differential equation with constant coefficients.
7. Prove that

$$\int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(2a-x) = f(x) \\ 0 & \text{if } f(2a-x) = -f(x) \end{cases}$$

Section-C

(Long answer type questions)

Answer any **three** questions of the following :

5×3=15

8. If $y = e^{a \sin^{-1} x}$, prove that

$$(1 - x^2)y_2 - xy_1 - a^2y = 0.$$

9. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

10. Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x}$ or

$$\int_0^{\frac{\pi}{2}} \log \sin x \, dx$$

11. Apply Maclaurin's theorem to prove that

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^n \frac{x^n}{n}$$

12. Solve any **one** of the following :

(i) $\frac{dy}{dx} + 1 = e^{x-y}$

(ii) $\cos^2 x - \frac{dy}{dx} + y = \tan x$

13. Solve any **one** of the following :

$$(i) \quad x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = x^4$$

$$(ii) \quad \frac{d^2 y}{dx^2} + y = \sin 2x$$

_____ x _____